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License No.: 4864

Protocol generation date:
No.:

LABORATORY TEST PROTOCOL

Patient code:
Age:
Sex:

Sample type: dried blood spot (DBS)
Sample collected:
Sample received:
Analysis date:

Test *	Result, measurement units	Interpretation	Reference range
Total Testosterone	nmol/l		9.16-31.8
SHBG	nmol/l		7-70
Free Testosterone **	pmol/l		184-749
DHEAS	μmol/l		(< 50 y.o.) 0.66 – 12.77 (> 50 y.o.) 0.54 – 10.14
17-OH-progesterone	nmol/l		0.87-6.24
Androstenedione	nmol/l		1.53-8.28
Cortisol	nmol/l		136.3-505.4
Cortisone	nmol/l		24.9-106.2
Cortisol/Cortisone ratio			3.32-12
Cortisol/DHEAS ratio			0.02-0.14
Testosterone/ Androstenedione ratio			>1.0

* Analytical method: SHBG was quantified by ELISA (5), whereas all other biomarkers were measured by LC-MS/MS (3)
** Calculated according to the Vermeulen equation

Employee license No.:

Understanding your Testosterone+ test results

This report analyzes a detailed panel of hormones and related biomarkers, including testosterone, DHEAS, cortisol, cortisone, SHBG, 17-OH-progesterone, androstenedione, and their key physiological relationships. By assessing both individual hormone levels and the balance between them, this report provides a broader view of your endocrine profile.

For simplicity, the biomarkers can be grouped into four key areas:

Area of Assessment	Biomarkers	What does it help evaluate?
Testosterone status & availability	Total Testosterone, SHBG, Free Testosterone	Overall testosterone status and the amount of biologically active testosterone available to tissues
Steroid hormone synthesis & androgen precursors markers	DHEAS, 17-OH-Progesterone, Androstenedione	Steroid hormone synthesis pathways, androgen precursors, and the potential contribution of the adrenal glands and testes to hormone production
Cortisol metabolism & stress response biomarkers	Cortisol, Cortisone	Cortisol activity, metabolism, and physiological stress-response processes
Integrated hormone ratios	Cortisol/Cortisone Ratio, Cortisol/DHEAS Ratio, Testosterone/Androstenedione Ratio	Relationships between key hormones and additional insight into overall hormonal balance and regulation

Each biomarker is described below, together with its relevance to endocrine health.

TESTOSTERONE STATUS & AVAILABILITY

TOTAL TESTOSTERONE

Overall testosterone status

Provides an overview of total circulating testosterone levels, including both the biologically active (free) fraction and the portion bound to transport proteins (SHBG and serum albumin) [1].

Optimal testosterone levels support healthy libido and sexual function, muscle mass, strength and recovery, energy production and physical performance, motivation, psychological well-being, bone density, normal sperm production, and reproductive function [2].

Low testosterone levels are associated with the opposite effects, including reduced libido, lower energy levels, fatigue, decreased muscle mass and strength, slower recovery from exercise, mood changes, reduced motivation, increased body fat accumulation [3], impaired fertility potential, and an increased risk of cardiometabolic dysfunction and cardiovascular events [4].

Higher testosterone levels may occur naturally in some individuals but can also reflect hormonal imbalance or external testosterone use (steroids/TRT). Elevated levels may be associated with increased acne, accelerated hair loss (called androgenic alopecia) in genetically susceptible individuals, reduced sperm count/infertility, prostate enlargement, mood changes (aggression, irritability), polycythemia (increased red blood cells) [5], and worsening sleep apnea [2].

SHBG (SEX HORMONE-BINDING GLOBULIN)

Regulation of hormone availability

By binding to testosterone and other sex hormones, SHBG helps determine how much of the hormone remains biologically active and available to tissues [6].

Higher SHBG levels may reduce the amount of free, active testosterone, whereas lower SHBG levels may increase its availability. This means that changes in SHBG can influence free testosterone levels even when total testosterone is within the reference range [6,7].

SHBG concentrations can be influenced by age, body composition, metabolic health, thyroid function, liver health, hormonal status, nutrition, and certain medications. For example, estrogens and thyroid hormones may increase SHBG levels, while androgens, obesity, insulin resistance, and some metabolic conditions may lower them.

FREE TESTOSTERONE

Biologically active testosterone available to tissues

Free testosterone can directly interact with cells and is therefore considered a better indicator of androgen activity than total testosterone alone. In this report, it is calculated from total testosterone and SHBG using the Vermeulen equation, with a standard albumin concentration used in the calculation.

The potential effects of low or high free testosterone levels generally mirror those described in the Total Testosterone section.



STEROID HORMONE SYNTHESIS & ANDROGEN PRECURSORS MARKERS

DHEAS (DEHYDROEPIANDROSTERONE SULFATE)

Adrenal androgen production

It serves as an important precursor for the production of testosterone, estrogens, and other sex hormones, making it a useful marker of adrenal androgen production [8].

DHEAS levels naturally decline with age and are often used as a marker of age-related endocrine changes. Levels within the reference range are generally consistent with healthy adrenal androgen production.

Low DHEAS levels may reflect age-related decline, chronic illness, or reduced adrenal hormone production. They have also been associated with reduced physical function, lower stress resilience, mood changes, cognitive changes, and increased cardiometabolic risk [9,10].

High DHEAS levels may indicate increased adrenal androgen activity. Mild elevations may occur in younger individuals or be influenced by stress-related factors, medications, or supplements, whereas markedly elevated levels may warrant further evaluation for underlying adrenal disorders.

17-OH-PROGESTERONE (17-OHP)

Early-stage steroid precursor

17-OH-progesterone is an intermediate steroid hormone formed during earlier stages of steroid hormone synthesis. It is involved in the pathway leading to cortisol production. When 21-hydroxylase function is impaired, accumulated 17-OHP may be diverted into androgen synthesis pathways, contributing to increased androstenedione and testosterone [11].

Higher 17-OHP levels may indicate increased activity of adrenal steroid hormone pathways or altered steroid hormone synthesis. In adults, elevated levels may be relevant when evaluating androgen-related biomarkers or possible non-classic congenital adrenal hyperplasia [11]. Markedly elevated levels may warrant further clinical evaluation to exclude underlying adrenal enzyme dysfunction.

Lower 17-OHP levels are often of limited clinical significance when other related biomarkers are within the reference range. In men, lower levels may also be observed during exogenous testosterone use [12].

Emerging evidence suggests that 17-OHP may serve as an indirect marker of intratesticular testosterone, an important factor for sperm production and male fertility. When interpreted alongside testosterone, androstenedione, and other hormonal biomarkers, it may provide additional insight into testicular steroidogenesis and reproductive function [12–14].

ANDROSTENEDIONE

Later-stage androgen precursor

It serves as a precursor for testosterone and estrogens and provides insight into androgen production pathways [15].

Higher androstenedione levels may suggest increased androgen production, whereas lower levels may reflect reduced activity of androgen synthesis pathways. However, isolated low values often have limited clinical significance when other related biomarkers are within the reference range [16].

Androstenedione is most informative when interpreted alongside testosterone, free testosterone, DHEAS, SHBG, and 17-OH-progesterone. Together with DHEAS, it may help provide additional insight into whether changes in androgen activity are more likely adrenal or gonadal in origin [17].

CORTISOL METABOLISM & STRESS RESPONSE BIOMARKERS

CORTISOL

Stress-response activity

It plays an important role in regulating the stress response, energy metabolism, immune function, and daily physiological rhythms.

Optimal cortisol levels support healthy adaptation to stress and overall well-being by mobilizing energy through the regulation of blood glucose levels, regulating the metabolism of fats, proteins, and carbohydrates [18], modulating immune function and inflammatory responses, and supporting blood pressure regulation and cardiovascular function. Cortisol also helps maintain the sleep-wake cycle, with levels naturally peaking in the morning [19].

Elevated cortisol levels may be associated with physiological or psychological stress, poor sleep quality, increased appetite, abdominal weight gain, elevated blood sugar levels, mood changes, and impaired recovery.

Low cortisol levels may reflect reduced cortisol production or altered cortisol regulation. They may be associated with fatigue, weakness, low blood pressure, dizziness, reduced stress tolerance, and difficulty maintaining normal blood sugar levels [18].

CORTISONE

Cortisol metabolism and regulation

Cortisone is a less active form of cortisol and is part of the body's cortisol metabolism. The body continuously converts cortisol to cortisone and back to help regulate tissue exposure to active cortisol.

Higher cortisone levels may reflect increased cortisol inactivation, whereas lower cortisone levels may indicate reduced cortisol inactivation. Cortisone is most informative when interpreted together with cortisol and the Cortisol/Cortisone ratio [20].

INTEGRATED HORMONE RATIOS

CORTISOL/CORTISONE RATIO

Balance between active and inactive cortisol

This balance may reflect how 11β -hydroxysteroid dehydrogenase enzymes regulate cortisol activation and inactivation in tissues [20].

Optimal ratio values are generally consistent with healthy cortisol regulation, stress resilience, recovery, metabolic health, and overall well-being.

A higher ratio may indicate greater relative exposure to active cortisol and may be associated with acute or prolonged stress, inflammation, impaired recovery, sleep disturbances, higher blood pressure, and adverse metabolic health outcomes [21,22].

A lower ratio may indicate lower relative exposure to active cortisol and, particularly when cortisol levels are also low, may be consistent with reduced adrenal cortisol activity [23]. This may be associated with fatigue, reduced stress responsiveness, and a lower ability to respond to physical or emotional stress. Markedly low values should be interpreted in the clinical context, including medication, supplements, and dietary factors.

CORTISOL/DHEAS RATIO

Balance between stress and resilience

While cortisol helps coordinate the body's response to stress, DHEAS serves as a precursor to sex hormones and has been associated with anti-inflammatory, antioxidant, and neuroprotective effects [24]. Together, these hormones provide insight into how the body balances stress-related and protective physiological processes.

Optimal ratio values are associated with balanced stress regulation, healthy recovery, and overall hormonal well-being.

A higher ratio may indicate increased physiological stress relative to recovery capacity and may result from higher cortisol, lower DHEAS, or both. Because DHEAS naturally declines with age, this ratio may also increase with aging. A higher ratio has been associated with fatigue, impaired recovery, reduced resilience, and accelerated biological aging [25].

Lower cortisol/DHEAS ratio values may reflect reduced cortisol activity relative to DHEAS. In many cases, this may be consistent with a more favorable balance between stress-related and protective hormonal activity [25]. However, markedly low values may also suggest reduced stress responsiveness or altered adrenal regulation. Interpretation should always consider absolute cortisol and DHEAS concentrations, symptoms, medications, and overall clinical context.

TESTOSTERONE/ANDROSTENEDIONE RATIO

Hormone conversion efficiency

This ratio reflects the relationship between testosterone and its immediate precursor, androstenedione. It provides insight into the efficiency of the final stages of testosterone synthesis and androgen conversion pathways.

Lower than expected ratio values may suggest reduced conversion of androstenedione to testosterone and can be associated with lower androgen availability, reduced libido, impaired reproductive function, and other signs of androgen deficiency [26].

Hormone biomarkers provide useful information about endocrine patterns, but they are most meaningful when interpreted together rather than individually. Results should be reviewed in the context of symptoms, medications, lifestyle factors, sampling conditions, and laboratory reference intervals, as hormone levels can naturally vary.

Thank you for trusting us with your hormone health assessment.

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